JavaScript 1.7

Modules, Errors & Serialization

Print-first interview refresher | exports → failures → data contracts

Trace each boundary:
who owns the value?
who handles the failure?

1 | ESM: explicit names, shared bindings

Named export
import { name as local }
Pick an exported name.

Default export
import local from "./api.mjs"
Namespace includes default.

Module scope
Own names; strict mode
Top-level this is undefined.

Load → link → evaluate. Static imports are linked before the importing body runs.
A live import reads the current binding. A local snapshot stays unchanged. Imports cannot be reassigned.

2 | Readiness, dynamic loading and build analysis

Cycles: legal graph; reading an uninitialized binding can throw ReferenceError.

import(): a promise for the module namespace; observe loading and evaluation failures.

Top-level await: delays dependent evaluation; does not block the entire process.

Tree shaking belongs to the bundler: track used exports AND preserve side effects.
A named import alone promises no bundle size. False sideEffects metadata can break initialization.

3 | CommonJS: legacy and interoperability

Node.js: .mjs = ESM; .cjs = CJS. Set the package type explicitly for .js.

require() returns module.exports. Reassigning exports only changes a local reference.

ESM → CJS: default is module.exports; detected named exports are not live bindings.

require(esm): tested Node 22.13.1 / 24.21.0 / 26.4.0 accept only graphs without TLA.

Conditional exports may select two implementations: test shared state and constructor identity.

4 | Throw → propagate → handle or rethrow

try execute the operation → **catch** handle, add context, or rethrow → **finally** release resources

A normal finally preserves the outcome; its own return or throw replaces it.

Add operation context with Error(message, { cause }). Preserve the original cause.

Await inside try to catch rejection there. A later callback is outside a completed try block.

5 | JSON carries a restricted data model

Object field: undefined / function / Symbol
Field omitted; root value → undefined.

Array: same values or a hole
Position becomes null; NaN → null too.

BigInt and cycles need an explicit representation. Dates usually become strings via toJSON.

toJSON → replacer. reviver visits children before the root; undefined deletes a property.

6 | Parse syntax, validate shape, then use data

JSON text syntax can fail → **Schema + version** valid JSON can still be invalid → **Explicit DTO** only agreed fields and types

Use strings for large IDs. Keep syntax, schema and module-loading failures distinguishable.

Interview check

Name the runtime. Trace readiness and causes. Specify the serialized format before writing code.